Contents

1 On manifolds with boundary

In this section we discuss how known theorems/lemmas fail for manifolds with non-emtpy boundary and how they can be generalized.

1.1 Regular values and transversality

It is known that for the regular value theorem, and more generally statements about transversality, to still hold for manifolds with boundary one needs to impose additional conditions on the boundary behaviour of a smooth function $f\colon M \to N$. Here is another lemma, that fits into the same category. We call $q \in N$ a **boundary value** of f if $q \in f(\partial M)$.

Lemma 1.1. *Let M be a manifold with boundary and $f\colon M \to \mathbb{R}$ a smooth function. Furthermore, let $a \in \mathbb{R}$ be a regular value of f, that is not a boundary value. Then $M_a := f^{-1}(-\infty, a]$ is a submanifold with boundary*

$$\partial M_a = f^{-1}(a) \cup (M_a \cap \partial M).$$

Proof sketch. For $p \in f^{-1}(a)$ we have $p \in \mathrm{int}(M)$ by assumption. Since $\mathrm{int}(M)$ is a boundaryless manifold (and an open subset of M) the local submersion theorem holds, i.e. f looks like the orthogonal projection $\mathbb{R}^n \to \mathbb{R}$ around p. Then the proof goes essentially the same as in the boundaryless case. $\square$

Counterexample 1.2. To see that Lemma 1.1 fails without the assumption that a is not a boundary value consider $M := S^1 \times [0, 1]$ and $f\colon M \to \mathbb{R}, (z, t) \mapsto t$. Then f is everywhere regular but $f^{-1}(-\infty, 0]$ does not satisfy the statements in Lemma 1.1.

Let M, N be manifolds with boundary and $f\colon M \to N$ smooth. We define the **boundary map** ∂f of f by

$$\partial f := f|_{\partial M}.$$

We now generalize two theorems from [GP74] about transversal maps by allowing the codomain to have boundary as well.

Theorem 1.3 (Transversality Theorem)**.** *Let M, N be manifolds with boundary, $f\colon M \to N$ smooth and $X \subseteq N$ a submanifold without boundary. If*

$$f \pitchfork X \text{ and } \partial f \pitchfork X$$

then $X' := f^{-1}(X)$ is a submanifold with boundary

$$\partial X' = X' \cap \partial M$$

and

$$\mathrm{codim}(X') = \mathrm{codim}(X).$$

Proof. The case $\partial N = \emptyset$ is shown in [GP74] (p. 60). If $\partial N \neq \emptyset$ consider the inclusion

$$\iota\colon N \to \mathrm{D}(N),$$

where $\mathrm{D}(N)$ denotes the double of N and define $F := \iota \circ f$. Then $Y := \iota(X)$ is a submanifold of $\mathrm{D}(N)$ and $X' = F^{-1}(Y)$. Furthermore, since ι is an embedding (and $\dim(N) = \dim(\mathrm{D}(N))$) it follows that

$$F \pitchfork Y \text{ and } \partial F \pitchfork Y,$$

such that the statement follows from the boundaryless case. $\square$

Theorem 1.4 (Parametric Transversality Theorem)**.** *Let M, N be manifolds with boundary and $X \subseteq N$ a submanifold without boundary. Furthermore, let S be a manifold without boundary and $F\colon M \times S \to N$ a smooth map with*

$$F \pitchfork X \textit{ and } \partial F \pitchfork X.$$

Then for almost every $s \in S$ the map $f_s\colon M \to N$ defined by $f_s(p) := F(p, s)$ satisfies

$$f_s \pitchfork X \textit{ and } \partial f_s \pitchfork X.$$

Proof. By Theorem 1.3 we know that

$$X' := F^{-1}(X)$$

is a submanifold (with boundary) of $M \times S$. Consider the projection $\pi\colon M \times S \to S$. We can then show, that for a regular value $s \in S$ of the map $\pi|_{X'}$ the maps f_s and ∂f_s are transversal to X, which is just some linear algebra. The proof is exactly the same as in the case $\partial N = \emptyset$, see [GP74] (p. 68). Then the statement follows from Sard‘s Theorem. $\square$

Theorem 1.5 (Transversality Homotopy Theorem)**.** *Let M, N be manifolds with boundary, $f\colon M \to N$ smooth and $X \subseteq N$ a submanifold without boundary. Then f is smoothly homotopic to a map $g\colon M \to N$ with*

$$g \pitchfork X \textit{ and } \partial g \pitchfork X.$$

Proof. First we may assume that $\mathrm{im}(f) \subseteq \mathrm{int}(N)$, since f is always smoothly homotopic to such a map (the inclusion $\mathrm{int}(N) \hookrightarrow N$ is a smooth homotopy equivalence). Define

$$Y := X \cap \mathrm{int}(N)$$

and

$$f'\colon M \to \mathrm{int}(N),\ p \mapsto f(p).$$

Using the Transversality Homotopy Theorem for the case $\partial N = \emptyset$ (see [GP74] p. 70) we get that f' is smoothly homotopic to a map $g'\colon M \to \mathrm{int}(N)$, such that

$$g' \pitchfork Y \text{ and } \partial g' \pitchfork Y.$$

Now define $g := \iota \circ f'$, where $\iota \colon \mathrm{int}(N) \hookrightarrow N$ is the inclusion. Since g does not intersect X at boundary points, i.e. $\mathrm{im}(g) \cap X \subseteq \mathrm{int}(N)$, the transversality properties of g' directly imply

$$g \pitchfork X \text{ and } \partial g \pitchfork X.$$

□

1.2 Morse Theory

For a smooth function $f \colon M \to \mathbb{R}$ and a regular value $a \in \mathbb{R}$ that is not a boundary value we define the **sublevel set** of a to be

$$M_a := f^{-1}(-\infty, a].$$

Lemma 1.1 guarantees, that M_a is a (sub-)manifold with boundary.

Lemma 1.6. *Let M be a compact manifold with boundary and $f \colon M \to \mathbb{R}$ a Morse function. Furthermore let $a, b \in \mathbb{R}$ be regular values of f, such that $[a, b]$ does not contain any critical values nor any boundary values. Then*

$$M_a \cong M_b.$$

Proof. Let $K := f^{-1}[a, b]$ and let $U := \{p \in \mathrm{int}(M) \mid p \text{ is a regular point of } f\}$. Then $U \subseteq M$ is open and $K \subseteq U$. Choose a cutoff function $\phi \colon M \to \mathbb{R}$ with

- $\phi|_K = 1$
- $\mathrm{supp}(\phi) \subseteq U$
- $0 \leq \phi \leq 1$

Now choose a pseudo gradient[1] X of f and define the smooth vector field

$$Y := \phi \frac{-1}{X \cdot f} X,$$

which we define to be 0 outside of U. Since Y vanishes in a neighbourhood around ∂M (or more generally since Y is tangent to ∂M, see [Lee11] Theorem 9.34) we can consider the flow of Y:

$$\theta \colon M \times \mathbb{R} \to M.$$

Since M is compact (and Y tangent to ∂M) each integral curve is defined for all time. Let $\theta_t := \theta(-, t) \colon M \to M$. We claim that

$$\theta_{b-a}(M_b) \subseteq M_a, \tag{1}$$
$$\theta_{a-b}(M_a) \subseteq M_b. \tag{2}$$

[1]Here we agree to the convention that $X \cdot f < 0$ outisde of critical points.

Let $p \in f^{-1}[a,b]$ and let $\gamma_p\colon \mathbb{R} \to M$ be the integral curve of Y starting at p. Consider

$$h\colon \mathbb{R} \to \mathbb{R},\ t \mapsto f(\gamma_p(t)).$$

Then for $t \in \mathbb{R}$ with $h(t) \in [a,b]$ we have $\phi(\gamma_p(t)) = 1$ and thus

$$h'(t) = \dot{\gamma_p}(t) \cdot f = Y_{\gamma_p(t)} \cdot f = -1.$$

So h is a smooth function with $c := h(0) = f(p) \in [a,b]$ and $h'(t) = -1$ as long as $h(t) \in [a,b]$. We claim that $h(c-a) = a$. To see this let

$$s := \sup\{t \in [0, c-a] \mid h([0,t]) \subseteq [a,b]\}.$$

Assume $s < c - a$. For continuity reasons we get that $h([0,s]) \subseteq [a,b]$ and thus $h'(t) = -1$ for $t \in [0,s]$. Then $h(t) = c - t$ for $t \in [0,s]$. But then $h(s) \in (a,b]$ and h is strictly decreasing around $t = s$ such that there exists $\varepsilon > 0$ with $h([0, s+\varepsilon]) \subseteq [a,b]$, contradicting the definition of s. So $s = c - a$ and we get that $h(t) = c - t$ for $t \in [0, c-a]$. Thus $h(c-a) = a$.

As $h'(t) \leq 0$ for all $t \in \mathbb{R}$ we conclude

$$f(\theta_{b-a}(p)) = h(b-a) \leq h(c-a) = a,$$

thus $\theta_{b-a}(p) \in M_a$. For $p \in M_b \setminus f^{-1}[a,b] = \mathrm{int} M_a$ we also have

$$\theta_{b-a}(p) \in M_a$$

as f decreases along gradient flow lines of Y. This shows (1). We argue similiarly for (2). Let $p \in M_a$ and let $\gamma_p\colon \mathbb{R} \to M$ be the integral curve of Y starting at p. Consider

$$h\colon \mathbb{R} \to \mathbb{R},\ t \mapsto f(\gamma_p(-t)).$$

Then $h'(t) = \phi(\gamma_p(-t)) \leq 1$. Assume that $h(b-a) = f(\theta_{a-b}(p)) > b$. Then by the mean value theorem there exists some $\xi \in [0, b-a]$ with

$$h'(\xi) = \frac{h(b-a) - h(0)}{b-a} \geq \frac{h(b-a) - a}{b-a} > 1,$$

which is a contradiction. So $f(\theta_{a-b}(p)) \leq b$, showing (2). Thus θ_{b-a} restricts to a diffeomorphism $M_b \xrightarrow{\cong} M_a$ with inverse θ_{a-b}. □

Counterexample 1.7. Example 1.2 also shows, that we need to impose conditions on the boundary values in Lemma 1.6. To see that we also need that condition for Lemma 1.9 we may similiarly define f as the height function on the disjoint union of two cylinders which are slightly shifted in height. Similiar examples show, that Theorem 1.10 fails without the assumption on boundary values.

Remark 1.8. We can weaken the assumptions of Lemma 1.6. First of all f does not need to be a Morse function. We can then still construct a vector field X, such that $X \cdot f < 0$ outisde of critical points. Secondly, we may replace the assumption that M is compact by instead requiring $f^{-1}[a,b]$ to be compact, which is for example the case if f is proper. We then only have to modify the proof in the definition of U. Instead we define U to be a *compact* neighbourhood of $K := f^{-1}[a,b]$, i.e. $K \subseteq \text{int}U$, which does not contain any critical points nor boundary points. For example we may define U as a union of finitely many (here we need compactness of $f^{-1}[a,b]$) compact sets K_i not containing neither critical points nor boundary points such that $\text{int}K_i$ still cover K.

Similiarly we may proof the following, for which we can also weaken the assumptions as described in the remark above.

Lemma 1.9. *Let M be a compact manifold with boundary and $f\colon M \to \mathbb{R}$ a Morse function. Furthermore let $a, b \in \mathbb{R}$ be regular values of f, such that $[a,b]$ does not contain any critical values nor any boundary values. Then M_a is a (not necessarily smooth) deformation retract of M_b.*

Proof. We consider the flow θ as in Lemma 1.6 and define a retraction

$$r\colon M_b \to M_a,\ p \mapsto \begin{cases} \theta(f(p)-a,p), & f(p) \geq a \\ p, & f(p) \leq a \end{cases}$$

As we have shown above $\theta(f(p)-a,p) \in f^{-1}(a)$. It follows that r is continous and well-defined. To see that r is a deformation retract consider

$$H\colon M_b \times [0,1] \to M_b, (p,t) \mapsto \begin{cases} \theta(t(f(p)-a),p), & f(p) \geq a \\ p, & f(p) \leq a \end{cases}$$

Then $H(-,0) = id_{M_b}$ and $H(-,1) = r$. □

We would now also like to generalize the result, that the topology of M changes by attaching a λ-cell when crossing a critical point c_0 of index λ to compact manifolds with boundary. This statement can now be shown using the two lemmas above by requiring $f^{-1}[a,b]$ to only contain the critical point c_0 and no boundary points. The proof now works exactly the same as for the boundaryless case, see [Mil63]. Let us state the theorem for manifolds with boundary:

Theorem 1.10. *Let M be a compact manifold with boundary and $f\colon M \to \mathbb{R}$ a Morse function. Let $c_0 \in M$ be a critical point of f of index λ and $c := f(c_0)$ as well as $\varepsilon > 0$, such that $f^{-1}[c-\varepsilon, c+\varepsilon]$ contains no boundary points and no critical points besides c. Then for ε sufficiently small $M_{c+\varepsilon}$ has the homotopy type of $M_{c-\varepsilon}$ with a λ-cell attached. More precisely, there exists an attaching map*

$$\phi\colon S^{\lambda-1} = \partial D^\lambda \to \partial M_{c-\varepsilon}$$

and a homotopy equivalence

$$r\colon M_{c+\varepsilon} \to M_{c-\varepsilon} \cup_\phi D^\lambda$$

with homotopy inverse

$$\iota \colon M_{c-\varepsilon} \cup_\phi D^\lambda \to M_{c+\varepsilon},$$

such that $r \circ \iota = id_{M_{c-\varepsilon} \cup_\phi D^\lambda}$.

Proof Sketch. Following [Mil63] we first construct $F \colon M \to \mathbb{R}$, such that

$$F^{-1}(-\infty, c+\varepsilon] = f^{-1}(-\infty, c+\varepsilon] = M_{c+\varepsilon}.$$

Furthermore $F^{-1}[c-\varepsilon, c+\varepsilon]$ does not contain any critical points or boundary points, such that $F^{-1}(-\infty, c-\varepsilon]$ is a deformation retract of $F^{-1}(-\infty, c+\varepsilon]$ by Lemma 1.9. We then show that $M_{c-\varepsilon} \cup_\phi D^\lambda$ is a deformation retract of $F^{-1}(-\infty, c-\varepsilon]$ (or more precisely the image of $M_{c-\varepsilon} \cup_\phi D^\lambda$ under an embedding is a deformation retract). This then shows that $M_{c-\varepsilon} \cup_\phi D^\lambda$ is a deformation retract of $M_{c+\varepsilon}$. □

However this does not yield a cell decomposition of M as we do not know how the topology changes when crossing boundary points. Instead we can use the fact that any compact manifold without boundary admits a cell-decomposition directly, to proof the case when M has non-empty boundary.

Theorem 1.11 (Cell-decompositions of Manifolds)**.** *Let M be compact manifold with or without boundary. Then M has the homotopy type of a CW-complex.*

Proof. The case when M has no boundary is shown in [Mil63]. If $\partial M \neq \emptyset$ we can argue as follows: consider the double $D(M)$ of M. Then $D(M)$ is a compact manifold without boundary. Now we construct a Morse function on $D(M)$. Choose a Morse function $f \colon M \to \mathbb{R}$ with $f \leq 0$ and $f(p) = 0$ if and only if $p \in \partial M$, such that f has no critical points on the boundary. Such Morse functions always exist, see [Vin24]. We can then glue f and $-f$ together to get a Morse function F on $D(M)$, again see [Vin24]. Then $M \cong F^{-1}(-\infty, 0]$ and 0 is a regular value of F, allowing us to refer back to the boundaryless case, which tells us that every sublevel set M_a for a regular value $a \in \mathbb{R}$ has the homotopy type of a CW-complex. □

Remark 1.12. We may also formulate Theorem 1.10 and 1.11 for handle bodies instead of CW-complexes. The proof is essentially the same but instead of constructing a deformation retract $F^{-1}(-\infty, c-\varepsilon] \to M_{c-\varepsilon} \cup_\phi D^\lambda$ one constructs a homeomorphism $F^{-1}(-\infty, c-\varepsilon] \to M_{c-\varepsilon} \cup_\phi D^\lambda \times D^{n-\lambda}$.

1.3 Integral curves and flows

For manifolds with boundary we essentially define integral curves in the same way as for boundaryless manifolds, but allow the domain to be any non-trivial interval (i.e. not a single point) containing 0. In particular this interval may only be half-open or closed. In general, the existence of integral curves on manifolds with boundary fails as illustrated by the following example:

Counterexample 1.13. Consider the lower halfspace $M := \mathbb{R}^2_{y\leq 0}$ and the vector field $X := \frac{\partial}{\partial x} + 2x\frac{\partial}{\partial y}$. Then for $p = (0,0)$ the integral curve is only defined for $t = 0$, since on $\mathbb{R}^2$ it would be given by $\gamma(t) = (t, t^2)$ but $t^2 > 0$ for $t \neq 0$.

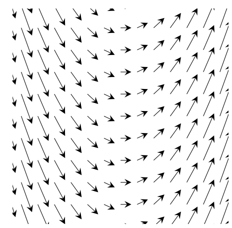

Figure 1: The vector field X. The integral curves are parabolas.

We do however still get a uniqueness result *if* the integral curve exists. Furthermore, if the vector field is sufficiently well-behaved we can guarantee the existence.

Lemma 1.14 (Uniqueness and existence of Integral curves)**.** *Let M be a manifold with boundary, X a smooth vector field on M and $p \in M$. If there exists an integral curve starting at p, then there exists a unique maximal integral curve $\gamma\colon I \to M$ starting at p, where one of the four following cases holds for $a \leq 0 \leq b$, $a < b$:*

(i) $I = (a, b)$

(ii) $I = [a, b)$ *and* $\gamma(a) \in \partial M$

(iii) $I = (a, b]$ *and* $\gamma(b) \in \partial M$

(iv) $I = [a, b]$ *and* $\gamma(a), \gamma(b) \in \partial M$

Furthermore, if X is either transversal to the boundary or everywhere tangent, the unique maximal integral curve starting at p always exists.

Proof sketch of Lemma 1.14. Let's begin by showing uniqueness. The proof is essentially the same as in the boundaryless case (see for example [Lee11]): We show that two integral curves $\gamma\colon I \to M$ and $\gamma'\colon I' \to M$ starting at p agree on their common domain. This then allows us to define the maximal integral curve on some interval I and every interval containing 0 is of the form $(i)-(iv)$ [2]. The only difference is that I and I' may not be open intervalls. So we have to show that uniqueness still applies. We claim: Let $U \subseteq \mathbb{R}^n$ be open, $F\colon U \to \mathbb{R}^n$ a smooth vector field, I an interval of the form $(i)-(iv)$ and $\gamma, \gamma'\colon I \to U$ integral curves both starting at $p \in U$. Then $\gamma = \gamma'$. If 0 is an interior point of I we can apply the usual uniqueness theorem for γ, γ' restricted to $\mathrm{int}(I)$. By continuity they must then also agree at the endpoint. So it remains to show uniqueness if 0 is a boundary point of I (say for example $I = [0, b)$). But by the existence theorem for integral curves we can find an integral curve $\eta\colon (-\varepsilon, \varepsilon) \to U$ starting at p and define

$$\beta\colon (-\varepsilon, b) \to U, t \mapsto \begin{cases} \eta(t), t \leq 0 \\ \gamma(t), t \geq 0 \end{cases}$$

Since $\eta(0) = \gamma(0)$ and $\dot{\eta}(0) = F(p) = \dot{\gamma}(0)$ we get that β is at least C^1. Similiarly we define

$$\beta'\colon (-\varepsilon, b) \to U, t \mapsto \begin{cases} \eta(t), t \leq 0 \\ \gamma'(t), t \geq 0 \end{cases}$$

By the uniqueness theorem we have

$$\beta = \beta'$$

and our claim follows. The case $I = (a, 0]$ goes analogous and for $I = [0, b]$ or $I = [a, 0]$ it is either trivial (if $a = b = 0$) or we consider $I \setminus \{b\}$ resp. $I \setminus \{a\}$ and apply the above argument. Equality at $t = a, t = b$ respectively then again follows by continuity.

To see that $\gamma(t) \in \partial M$ for $t \in \{a, b\}$ a boundary point of I we note that for $p = \gamma(t) \in \operatorname{int}(M)$ we can define the integral curve starting at p on some open interval $(-\varepsilon, \varepsilon)$ by the existence theorem for integral curves (Picard-Lindelöff). But then t is not a boundary point of I since I is maximal.

Next we show the existence if X is transversal to ∂M. For $p \in \operatorname{int}(M)$ this follows directly from the Picard-Lindelöff Theorem as in the boundaryless case. If $p \in \partial M$, we may argue similiarly: Consider a chart $h\colon \mathbb{R}^n_{x_n \geq 0} \supseteq U \to M$ around $p = h(0)$ and let Y be the vector field X in this chart, i.e. $D_q h(Y_q) = X_{h(q)}$. Then by definition of smoothness we can extend Y to a smooth vector field Y' on some open neighbourhood $U' \subseteq \mathbb{R}^n$ around $0 = h^{-1}(p)$. Applying Picard-Lindelöff we get the existence of an integral curve $\gamma\colon (-\varepsilon, \varepsilon) \to U'$ of Y' starting at 0. Write

$$Y' = \sum_i F_i \frac{\partial}{\partial x_i}$$

and

$$\gamma = (\gamma_1, \ldots, \gamma_n).$$

Assume that X is inwards pointing along the boundary. After restricting to a smaller domain we may assume $F_n \geq 0$. Then

$$\gamma_n'(t) = F_n(\gamma(t)) \geq 0,$$

such that γ_n is increasing. In particular $\gamma_n(t) \geq 0$ for $t \in [0, \varepsilon)$, hence $\gamma(t) \in U$ for $t \in [0, \varepsilon)$. Then

$$\eta := h \circ \gamma\colon [0, \varepsilon) \to M$$

is the desired integral curve.

For the case that X is everywhere tangent to ∂M see [Lee11] (Theorem 9.34).

$\square$

On compact manifolds we can say more about the maximal Interval I:

Lemma 1.15 (Existence of Integral curves on compact manifolds)**.** *If M is compact in Lemma 1.14 then the maximal interval I has one of the following forms:*

(i) $I = (-\infty, \infty)$

(ii) $I = [a, \infty)$ *and* $\gamma(a) \in \partial M$

(iii) $I = (-\infty, b]$ *and* $\gamma(b) \in \partial M$

(iv) $I = [a, b]$ *and* $\gamma(a), \gamma(b) \in \partial M$

Proof. Let $\gamma \colon I \to M$ be the unique maximal integral curve guaranteed by Lemma 1.14. We will show that any open boundary point of I must be $\pm\infty$. For simplicity let us only consider $I = [a, b)$. The other cases are completely analogous. By compactness we can find a sequence $0 < t_n \nearrow b$ such that $\gamma(t_n)$ coverges to a point $q \in M$.
Case 1: $q \in \mathrm{int}(M)$**.** By Picard-Lindelöff we can find an open neighbourhood $U \subseteq M$ around q and $\varepsilon > 0$, such that for every point $x \in U$ the intgeral curve starting at x is defined for $t \in (-\varepsilon, \varepsilon)$. For some fixed m sufficiently large we have $q_0 := \gamma(t_m) \in U$. Assume that $b - t_m < \frac{\varepsilon}{2}$. But then we can extend γ to $[a, b + \frac{\varepsilon}{2})$ as follows. Consider

$$\gamma' \colon [a - t_m, b - t_m) \to M,\ t \mapsto \gamma(t + t_m).$$

Then γ' is a integral curve starting at q_0. Let $\alpha \colon (-\varepsilon, \varepsilon) \to M$ be another integral curve starting at q_0. Then α and γ' agree on their common domain by the arguments in Lemma 1.14 and thus we can put them together to define an integral curve

$$\beta \colon J \to M$$

for $J := [a - t_m, b - t_m) \cup (-\varepsilon, \varepsilon) \supseteq [a - t_m, \varepsilon)$. But then we can define

$$\gamma'' \colon [a, \varepsilon + t_m) \to M,\ t \mapsto \beta(t - t_m),$$

which is an integral curve starting at $\gamma(0) = p$. However

$$b + \frac{\varepsilon}{2} < t_m + \varepsilon$$

and so $[a, b + \frac{\varepsilon}{2}) \subseteq [a, \varepsilon + t_m)$, contradicting maximality of γ.

[2] If we put together integral curves defined on open intervals that agree on their common domain, it is clear that this results in a smooth curve, since around each t this curve is given by one of the smooth curves which we put together. If we put together integral curves γ_1, γ_2 defined on something like $I_1 = (a, b]$ and $I_2 = [b, c)$ we have to be more careful around the boundary point b. But the same argument as we applied to β in the proof of Lemma 1.14 shows that putting γ_1 and γ_2 together results in a C^1-curve. We can then apply the usual uniqueness theorem to see that this curve is already given by a C^∞-curve.

Case 2: $q \in \partial M$**.** We argue similiarly as above. In a local chart around q we can again apply Picard-Lindelöff (smoothness of X guarantees by definition the existence of a smooth extension of X to an open neighbourhood in $\mathbb{R}^n$). We then again take $q_0 := \gamma(t_m)$ sufficiently close to q, choose an integral curve α starting at q, which we can use to define an integral curve α' starting at q_0 defined on some open interval $(-\varepsilon, \varepsilon)$ by translation in time as above. Also we define γ' as above. The same argument as above then shows that we can extend γ' (in the local chart) to an integral curve β on $J := [a - t_m, b - t_m) \cup (-\varepsilon, \varepsilon) \supseteq [a - t_m, b - t_m]$. In particular we have $\beta([a - t_m, b - t_m]) \subseteq M$. Translating in time again yields an integral curve $\gamma'' \colon [a, b] \to M$ contradicting maximality. $\square$

Remark 1.16 (A note on flows for manifolds with boundary). In general the concept of a flow is not well behaved if M has boundary, as integral curves may not even exist (see Counterexample 1.13). In [Lee11] the author shows, that if X is tangent to ∂M we still get a well-behaved flow, i.e. it is defined on some open subset $\mathcal{O} \subseteq M \times \mathbb{R}$. I think that if we instead assume, that X is transversal to ∂M, we still get a "good" flow domain in the sense that we can define the flow on some subset $\mathcal{D} \subseteq M \times \mathbb{R}$, which is a codimension 0 submanifold with boundary and $I_p := \{t \in \mathbb{R} \mid (p, t) \in \mathcal{D}\}$ is an interval around 0 for each $p \in M$.

The statement in the above remark should follow from the following lemma, which allows us to define submanifold charts for $\mathcal{D}$.

Lemma 1.17. *Let M be a manifold with boundary and X a vector field of M, which is inward pointing along ∂M. Let*

$$M_\partial := \{p \in M \mid \mathrm{im}\gamma_p \cap \partial M \neq \emptyset\}$$

be all points in M whose trajectory contains boundary points (this is at most one for each $p \in M$). For $p \in M_\partial$ the integral curve γ_p starting at p is defined on some interval $[-\tau(p), a)$ for some $\tau(p) \geq 0$ and $a > -\tau(p)$. Then the following statements hold true

(i) M_∂ is an open subset of M

(ii) $\tau \colon M_\partial \to \mathbb{R}_{\geq 0}$, $p \mapsto \tau(p)$ is smooth

Proof sketch. The idea is to consider the maximal flow

$$\theta \colon \mathcal{D} \to M, \ (p, t) \mapsto \gamma_p(t)$$

on some open subset $\mathcal{D} \subseteq \partial M \times [0, \infty)$ and show that this is an embedding. Then

$$M_\partial = \mathrm{im}\theta$$

is open and

$$\tau = \pi \circ \theta^{-1} \colon M_\partial \to [0, \infty)$$

for $\pi \colon \partial M \times [0, \infty) \to [0, \infty)$ is the canonical projection map. $\square$

Lemma 1.18. *Injective Flowouts are already embeddings*

2 Cartan Calculus

2.1 Lie derivative and Cartan's magic formula

In this section we discuss two differential operators on the graded algebra $\Omega^\bullet(M)$ of differential forms on a manifold M, namely the Lie derivative $\mathcal{L}_X$ and the interior product ι_X, where X is a smooth vector field on M. The main goal of this section is to show well-definedness of $\mathcal{L}_X$ and proof the following theorem, describing the relationships between $\mathcal{L}_X, \iota_X$ and the exterior derivative d.

Theorem 2.1. *Let M be a manifold and $X, Y \in \mathscr{X}(M)$. Then the maps*

$$\begin{aligned} d &: \Omega^r(M) \to \Omega^{r+1}(M), \\ \mathcal{L}_X &: \Omega^r(M) \to \Omega^r(M), \\ \iota_X &: \Omega^r(M) \to \Omega^{r-1}(M) \end{aligned}$$

are differential operators satisfying the following Leibniz rules for $\alpha, \beta \in \Omega^\bullet(M)$:

$$\begin{aligned} d(\alpha \wedge \beta) &= d\alpha \wedge \beta + (-1)^{|\alpha|} \alpha \wedge d\beta, \\ \mathcal{L}_X(\alpha \wedge \beta) &= \mathcal{L}_X \alpha \wedge \beta + \alpha \wedge \mathcal{L}_X \beta, \\ \iota_X(\alpha \wedge \beta) &= \iota_X \alpha \wedge \beta + (-1)^{|\alpha|} \alpha \wedge \iota_X \beta, \end{aligned}$$

as well as the following identities:

(i) $d^2 = 0$,

(ii) $d\mathcal{L}_X = \mathcal{L}_X d$,

(iii) $\mathcal{L}_X = d\iota_X + \iota_X d$,

(iv) $\mathcal{L}_X \mathcal{L}_Y - \mathcal{L}_Y \mathcal{L}_X = \mathcal{L}_{[X,Y]}$,

(v) $\mathcal{L}_X \iota_Y - \iota_Y \mathcal{L}_X = \iota_{[X,Y]}$,

(vi) $\iota_X \iota_Y + \iota_Y \iota_X = 0$.

Firstly let us restate the definition of the Lie derivative. For a smooth manifold M without boundary, $\alpha \in \Omega^r(M)$ and $X \in \mathscr{X}(M)$ we define

$$(\mathcal{L}_X \alpha)_p := \frac{d}{dt}\bigg|_{t=0} (\phi_t^* \alpha)_p,$$

where $\phi_t = \phi(t, -)$ denotes the flow of X.

Lemma 2.2. $(\mathcal{L}_X \alpha)_p$ *exists for all $p \in M$ and defines a smooth r-form on M.*

Before showing the lemma, let us introduce the notion of a smooth family of r-forms.

Definition 2.3 (Smooth family of r-forms)**.** Let β_t be a family of r-forms on M indexed over some open Interval I. We call β_t smooth if in some (then all) local coordinates $x^1, \ldots, x^n$ we can write

$$(\beta_t)_p = \sum \beta^{i_1,\ldots,i_r}(t,p) d_p x^{i_1} \wedge \ldots \wedge d_p x^{i_r}$$

for smooth functions $\beta^{i_1,\ldots,i_r}$. If I is closed we call β_t smooth if it can be extended to a smooth family β'_t indexed over some open open interval $I' \supseteq I$.

Then $t \mapsto (\beta_t)_p$ is a smooth curve in the vector space $\mathrm{Alt}^r(T_pM)$, such that

$$\left(\frac{d}{dt}\bigg|_{t=t_0} \beta_t \right)_p := \frac{d}{dt}\bigg|_{t=t_0} (\beta_t)_p$$

is a well-defined r-form. Furthermore, in local coordinates we have

$$\frac{d}{dt}\bigg|_{t=t_0} \beta_t = \sum \frac{\partial \beta^{i_1,\ldots,i_r}}{\partial t}(t_0, -) dx^{i_1} \wedge \ldots \wedge dx^{i_r},$$

such that this r-form is also smooth and moreover depends smoothly on t_0. So

$$\dot{\beta}_t := \frac{d}{dt}\beta_t$$

defines a smooth family of r-forms by

$$\left(\frac{d}{dt}\beta_t \right)_{t_0} := \frac{d}{dt}\bigg|_{t=t_0} \beta_t.$$

Note that if $\partial M = \emptyset$ we can also directly allow for closed intervals I in the definition above. However we may then always extend the family β_t to an open interval (in fact to all of $\mathbb{R}$), as β_t can equivalently be considered a smooth r-form β on $M \times I \subseteq M \times \mathbb{R}$. Under this correspondence we have the following interesting relationship (without proof):

$$\dot{\beta}_t = \mathcal{L}_{\frac{\partial}{\partial t}} \beta.$$

Proof of Lemma 2.2. Let $x_1, \ldots, x_n$ be local coordinates on some open neighbourhood $U \subseteq M$ of p. Choose a smaller neighbourhood $U' \subseteq U$ of p and $\varepsilon > 0$, such that the flow

$$\phi\colon (-\varepsilon, \varepsilon) \times U' \to U$$

of X is defined. Write

$$\alpha = \sum \alpha^{i_1,\ldots,i_r} dx^{i_1} \wedge \ldots \wedge dx^{i_r}.$$

We compute:

$$\begin{aligned}(\phi_t^*\alpha)_p &= (d_p\phi_t)^* \left(\sum \alpha^{i_1,\ldots,i_r}(\phi_t(p))\, d_{\phi_t(p)} x^{i_1} \wedge \ldots \wedge d_{\phi_t(p)} x^{i_r} \right) \\ &= \sum \alpha^{i_1,\ldots,i_r}(\phi_t(p))\, d_p(x^{i_1} \circ \phi_t) \wedge \ldots \wedge d_p(x^{i_r} \circ \phi_t)\end{aligned}$$

Expanding the differentials

$$d_p(x^{i_k} \circ \phi_t) = \sum_{j=1}^{n} \frac{\partial(x^{i_k} \circ \phi_t)}{\partial x_j}(p) d_p x_j$$

we see that $\beta_t := \phi_t^* \alpha$ is a smooth family of r-forms on U' indexed over $t \in (-\varepsilon, \varepsilon)$. Thus

$$(\mathcal{L}_X \alpha)_p = \left(\frac{d}{dt}\bigg|_{t=0} \beta_t \right)_p$$

exists and depends smoothly on p.

□

Proposition 2.4 (Basic properties of the Lie derivative)**.** *Let* $\alpha \in \Omega^r(M), \beta \in \Omega^s(M)$*,* $\lambda \in \mathbb{R}$ *and* $X \in \mathscr{X}(M)$*. Then the Lie derivative satisfies the following properties:*

(i) Locality: $(\mathcal{L}_X \alpha)_U = \mathcal{L}_{X|_U} \alpha|_U$ *for* $U \subseteq M$ *open*

(ii) Linearity in α*:* $\mathcal{L}_X(\lambda \alpha + \beta) = \lambda \mathcal{L}_X \alpha + \mathcal{L}_X \beta$

(iii) Leibniz rule: $\mathcal{L}_X(\alpha \wedge \beta) = \mathcal{L}_X \alpha \wedge \beta + \alpha \wedge \mathcal{L}_X \beta$

(iv) Compatibility with d*:* $\mathcal{L}_X(d\alpha) = d(\mathcal{L}_X \alpha)$

Proof. (i) follows from the fact that locally both sides are given by

$$\frac{d}{dt}\bigg|_{t=0} \beta_t$$

for the smooth family β_t defined in the proof of Lemma 2.2. Moreover, (ii) follows immediately from the definition. To see (iii) write

$$\mathcal{L}_X(\alpha \wedge \beta)_p = \frac{d}{dt}\bigg|_{t=0} (\phi_t^* \alpha)_p \wedge (\phi_t^* \beta)_p.$$

We claim: For a vector space V and smooth curves $\alpha_t \in \mathrm{Alt}^r(V), \beta_t \in \mathrm{Alt}^s(V)$ the following holds:

$$\frac{d}{dt}\bigg|_{t=0} (\alpha_t \wedge \beta_t) = \left(\frac{d}{dt}\bigg|_{t=0} \alpha_t \right) \wedge \beta_0 + \alpha_0 \wedge \left(\frac{d}{dt}\bigg|_{t=0} \beta_t \right)$$

W.l.o.g. let

$$\alpha_t = a(t)\alpha', \ \beta_t = b(t)\beta'$$

for some $\alpha' \in \mathrm{Alt}^r(V), \beta' \in \mathrm{Alt}^s(V)$ and smooth functions a, b. Then the general case follows from linearity. We compute:

$$\begin{aligned}
\left.\frac{d}{dt}\right|_{t=0} (\alpha_t \wedge \beta_t) &= \left.\frac{d}{dt}\right|_{t=0} a(t)b(t)\alpha' \wedge \beta' \\
&= \left(\frac{da}{dt}(0)b(0) + a(0)\frac{db}{dt}(0)\right)\alpha' \wedge \beta' \\
&= \frac{da}{dt}(0)\alpha' \wedge b(0)\beta' + a(0)\alpha' \wedge \frac{db}{dt}(0)\beta' \\
&= \left(\left.\frac{d}{dt}\right|_{t=0} \alpha_t\right) \wedge \beta_0 + \alpha_0 \wedge \left(\left.\frac{d}{dt}\right|_{t=0} \beta_t\right).
\end{aligned}$$

This shows the claim and thus (iii). To see (iv) let β_t be again the smooth family of r-forms defined in the proof of Lemma 2.2 and write in local coordinates $x_1, \ldots, x_n$:

$$\beta_t = \sum \beta^{i_1,\ldots,i_r}(t,-)dx^{i_1} \wedge \ldots \wedge dx^{i_r}.$$

By linearity we may assume that all functions $\beta^{i_1,\ldots,i_r}$ except for one vanish, i.e.

$$\beta_t = f(t,-)dx^{j_1} \wedge \ldots \wedge dx^{j_r}$$

for some smooth function f and $1 \leq j_1 < \cdots < j_r \leq n$. Then

$$d\beta_t = \sum_{k=1}^{n} \frac{\partial f(t,-)}{\partial x_k} dx^k \wedge dx^{j_1} \wedge \ldots \wedge dx^{j_r}.$$

Thus

$$\begin{aligned}
\mathcal{L}_X(d\alpha) &= \left.\frac{d}{dt}\right|_{t=0} \phi_t^* d\alpha = \left.\frac{d}{dt}\right|_{t=0} d\beta_t \\
&= \sum_{k=1}^{n} \frac{\partial^2 f}{\partial t \partial x_k}(0,-)dx^k \wedge dx^{j_1} \wedge \ldots \wedge dx^{j_r} \\
&= d\left(\frac{\partial f}{\partial t}(0,-)dx^{j_1} \wedge \ldots \wedge dx^{j_r}\right) \\
&= d(\left.\frac{d}{dt}\right|_{t=0} \beta_t) = d(\mathcal{L}_X \alpha)
\end{aligned}$$

□

For convenience, let us also restate the definition of ι_X

Definition 2.5. Let $X \in \mathscr{X}(M)$ and $\alpha \in \Omega^r(M)$. For $r > 0$ define

$$\iota_X \alpha := \alpha(X, -) \in \Omega^{r-1}(M)$$

For $r = 0$ let $\iota_X \alpha := 0$.

Clearly ι_X is linear and only depends on the local behaviour of X and α. Furthermore, it is easily seen that

$$\iota_X(\alpha \wedge \beta) = \iota_X\alpha \wedge \beta + (-1)^{|\alpha|}\alpha \wedge \iota_X\beta.$$

Thus far the Lie derivative has only been defined on manfiolds without boundary. To extend the definition of $\mathcal{L}_X\alpha$ to manifolds with boundary, first choose extensions X' and α' of X and α, respectively, to the double of M and then define

$$\mathcal{L}_X\alpha := (\mathcal{L}_{X'}\alpha')|_M$$

Of course it is not clear a priori that this definition does not depend on the choice of X' and α'. However, the following proposition, which is one of the formulas of Theorem 2.1, shows that it is in fact independent.

Proposition 2.6 (Cartan's magic formula)**.** *Let $X \in \mathscr{X}(M)$. Then the following identity holds:*

$$\mathcal{L}_X = d\iota_X + \iota_X d$$

Proof. Let $\mathcal{D}_X := d\iota_X + \iota_X d$. From the discussion above it is easy to verify:

$$\mathcal{D}_X(\alpha \wedge \beta) = \mathcal{D}_X\alpha \wedge \beta + \alpha \wedge \mathcal{D}_X\beta.$$

Moreover, $\mathcal{D}_X$ clearly satisfies locality, linearity and commutes with d. Thus we can establish

$$\mathcal{D}_X = \mathcal{L}_X$$

by showing equality on $\Omega^0(M)$. So let $f \in \Omega^0(M)$, then

$$\mathcal{D}_X f = \iota_X df = X \cdot f$$

and

$$\mathcal{L}_X f = \left.\frac{d}{dt}\right|_{t=0} (f \circ \phi_t) = df(X) = X \cdot f$$

This shows the proposition. $\square$

Now let us proof the remaining formulas of Theorem 2.1, which are shown similiarly.

Proof of Theorem 2.1. Equation (i) is only mentioned for the sake of completeness and will not be shown here. See for example [Ste23] for a proof. Equations (ii) and (iii) have already been shown above.

To see (iv) note that again both sides of the equation are differential operators satisfying the same Leibniz rule and commute with d, such that it suffices to show equality on $\Omega^0(M)$. But this follows immediately from the definition of the Lie bracket.

We argue similiarly for (v), but in this equation the differential operators on each side do not commute with d. However they still satisfy the same Leibniz rule, so we only need to show equality on $\Omega^0(M)$ as well as on the exact one-forms. Clearly both sides vanish on $\Omega^0(M)$ (by definition of the interior product) and for $\alpha = df \in \Omega^1(M)$ we compute:

$$\begin{aligned}(\mathcal{L}_X\iota_Y - \iota_Y\mathcal{L}_X)(df) &= \mathcal{L}_X(Y\cdot f) - (\iota_Y d\mathcal{L}_X)(f)\\ &= XYf - YXf\\ &= [X,Y]\cdot f = \iota_{[X,Y]}(df).\end{aligned}$$

Finally, (vi) is just a way to rewrite

$$\alpha(X,Y,-) = -\alpha(Y,X,-).$$

□

As an applictaion of Theorem 2.1 we can show the following identity:

Corollary 2.7. *For $\alpha \in \Omega^1(M)$ and $X, Y \in \mathscr{X}(M)$ the following identity holds:*

$$d\alpha(X,Y) = X\cdot\alpha(Y) - Y\cdot\alpha(X) - \alpha([X,Y])$$

Proof. By Theorem 2.1 (iii) we have:

$$(\iota_Y\mathcal{L}_X)(\alpha) = Y\cdot\alpha(X) + d\alpha(X,Y).$$

Moreover, applying (v) yields:

$$(\iota_Y\mathcal{L}_X)(\alpha) = X\cdot\alpha(Y) - \alpha([X,Y]).$$

The corollary follows. □

2.2 Moser‘s trick

Let M be a manifold (without boundary). Given a smooth family α_t of differential forms on M indexed over $t \in [0,1]$ we want to find a family of diffeomorphisms $\varphi_t\colon M \to M$ (i.e. an isotopy), such that $\varphi_0 = id_M$ and

$$\varphi_t^*\alpha_t = \alpha_0 \;\text{ for all } t \in [0,1],$$

or equivalently:

$$\frac{d}{dt}\varphi_t^*\alpha_t = 0. \tag{3}$$

Directly constructing such an isotopy is difficult but we can make use of the following trick: The isotopy φ_t corresponds to the level preserving diffeomorphism

$$\Phi\colon M\times[0,1] \to M\times[0,1], (p,t) \mapsto (\varphi_t(p), t).$$

This diffeomorphism then defines a vector field $\Phi_*\frac{\partial}{\partial t}$ on $M\times[0,1]$, whose integral curves uniquely determine φ_t. This suggests that in order to construct an isotopy φ_t we can construct a vector field Y on $M\times[0,1]$, of the form

$$Y_{(p,t)} = ((X_t)_p, \frac{\partial}{\partial t})$$

for a time dependent vector field X_t on M (this trick is also explained in [Mil65] Thm. 5.6).

Then (3) can be reformulated as a condition on X_t, which is shown in the following Lemma.

Lemma 2.8. *In the situation above the following holds:*

$$\frac{d}{dt}\varphi_t^*\alpha_t = \phi_t^*(L_{X_t}\alpha_t + \dot{\alpha}_t)$$

Proof. Assume that α_t is defined for all $t \in \mathbb{R}$ and consider the smooth two-parameter family $\phi_t^*\alpha_s$ for $(t,s) \in [0,1] \times \mathbb{R}$. Then for each $p \in M$ we have a smooth map

$$g\colon [0,1] \times \mathbb{R} \to T_p^*M,\ (t,s) \mapsto (\phi_t^*\alpha_s)_p$$

and

$$\left.\frac{d}{dt}\right|_{t=t_0} (\varphi_t^*\alpha_t)_p = \frac{\partial g}{\partial t}(t_0,t_0) + \frac{\partial g}{\partial s}(t_0,t_0)$$

By Proposition 2.9 we have:

$$\left.\frac{\partial}{\partial t}\right|_{t=t_0} \varphi_t^*\alpha_s = \varphi_{t_0}^*(\mathcal{L}_{X_{t_0}}\alpha_s)$$

Moreover, by writing α_s in local coordinates it is easily seen that

$$\left.\frac{\partial}{\partial s}\right|_{s=t_0} \varphi_t^*\alpha_s = \varphi_t^*\left(\left.\frac{\partial}{\partial s}\right|_{s=t_0} \alpha_s\right) = \varphi_t^*\dot{\alpha}_{t_0}.$$

This completes the proof. ☐

So (3) can be reformulated to the condition:

$$0 = L_{X_t}\alpha_t + \dot{\alpha}_t = d\iota_{X_t}\alpha_t + \iota_{X_t}d\alpha_t + \dot{\alpha}_t,$$

where we used Theorem 2.1 (iii) in the second equation. Hence it suffices to construct a time dependent vector field X_t satisfying the above conditon and whose integral curves exist for time $t = 1$.

2.3 Further computation rules

The Lie derivative $\mathcal{L}_X\alpha$ was defined as

$$\left.\frac{d}{dt}\right|_{t=0} \phi_t^*\alpha,$$

where ϕ_t is the flow of X. We might also want to differentiate this expression at some other time $t = t_0$. The following proposition shows how this can be computed in terms of $\mathcal{L}_X\alpha$. In fact, we show a more general statement, where ϕ_t is the flow of a time dependent vector field.

Proposition 2.9. *Let M be a manifold without boundary, X_t a time dependent vector field on M and $\alpha \in \Omega^r(M)$. Denote by ϕ_t the (time-dependent) flow of X_t and consider the open set $M_{t_0} = \{p \in M \mid \phi_t(p) \text{ exists for } t = t_0\}$. Then*

$$\frac{d}{dt}\bigg|_{t=t_0} (\phi_t^*\alpha)_p = \phi_{t_0}^*(\mathcal{L}_{X_{t_0}}\alpha)_p$$

for all points $p \in M_{t_0}$.

Proof. If X_t does not depend on time, i.e. $X_t = X_0$ for all t where it is defined, the proposition follows from a simple computation:

$$\begin{aligned}\frac{d}{dt}\bigg|_{t=t_0} (\phi_t^*\alpha)_p &= \frac{d}{dt}\bigg|_{t=0} (\phi_{t+t_0}^*\alpha)_p = \frac{d}{dt}\bigg|_{t=0} (\phi_{t_0}^*\phi_t^*\alpha)_p \\ &= \phi_{t_0}^*\frac{d}{dt}\bigg|_{t=0} (\phi_t^*\alpha)_p = \phi_{t_0}^*(\mathcal{L}_{X_0}\alpha)_p = \phi_{t_0}^*(\mathcal{L}_{X_{t_0}}\alpha)_p\end{aligned}$$

In the general case we argue as follows: The maps

$$\Omega^r(M_{t_0}) \ni \alpha \mapsto \frac{d}{dt}\bigg|_{t=t_0} \phi_t^*\alpha$$

and

$$\Omega^r(M_{t_0}) \ni \alpha \mapsto \phi_{t_0}^*\mathcal{L}_{X_{t_0}}\alpha$$

are both differential operators satisfying the same Leibniz rule and commute with d. For the second one this just follows from Proposition 2.4 and for the first one we can carry out precisely the same proof as of Proposition 2.4. Thus we can again establish equality by showing that they agree on $\Omega^0(M_{t_0})$. So for $f \in \Omega^0(M_{t_0})$ we compute:

$$\begin{aligned}\frac{d}{dt}\bigg|_{t=t_0} (\phi_t^*f)_p &= \frac{d}{dt}\bigg|_{t=t_0} f(\phi_t(p)) \\ &= d_{\phi_{t_0}(p)}f((X_{t_0})_{\phi_{t_0}(p)}) \\ &= (X_{t_0}\cdot f)(\phi_{t_0}(p)) \\ &= \phi_{t_0}^*(X_{t_0}\cdot f)(p) \\ &= \phi_{t_0}^*(\mathcal{L}_{X_{t_0}}f)_p\end{aligned}$$

This shows the statement. □

Here are two more rules for computing the time-derivative of a smooth family of $r-$forms, which are occasioanlly useful:

Lemma 2.10. *Let α_t be a smooth time-dependent family of differential forms on a manifold M and $f\colon N \to M$ a smooth map. Then the following holds:*

(i) $\frac{d}{dt}f^*\alpha_t = f^*\frac{d}{dt}\alpha_t$

(ii) $\frac{d}{dt} d\alpha_t = d(\frac{d}{dt}\alpha_t)$

Proof. Both statements can easily be verified by directly computing both sides of the equation in local coordinates. For *(ii)* we have in fact already seen this computation in the proof of Lemma 2.4. $\square$

A cooking recipe

The proof of Lemma 2.8 gives a cooking recipe for computing derivatives of smooth families of differential forms indexed over some time t, where t appears at multiple instances. Namely, fix t for all but one instance and differentiate. Then take the sum of all of the results. We illustrate this in an example.

Example 2.11. Let X_t be some time dependent vector field on a manifold M with (time-dependent) flow ϕ_t. Furthermore, let α_t be a smooth family of differential forms and f_t a smooth family of functions on M. We want to compute:

$$\frac{d}{dt}\phi_t^*(e^{f_t}\alpha_t).$$

Begin by fixing all but the first occurence of t and differentiate, i.e. compute

$$\begin{aligned}\frac{\partial}{\partial t}\phi_t^*(e^{f_s}\alpha_s) = \phi_t^*(\mathcal{L}_{X_t}(e^{f_s}\alpha_s)) &= \phi_t^*((X_t \cdot e^{f_s})\alpha_s + e^{f_s}\mathcal{L}_{X_t}\alpha_s) \\ &= \phi_t^*((X_t \cdot f_s)e^{f_s}\alpha_s + e^{f_s}\mathcal{L}_{X_t}\alpha_s)\end{aligned}$$

Here we used Proposition 2.9 and the Leibniz rule for the Lie derivative. Next fix all but the second occurence of t and using Lemma 2.10 compute:

$$\frac{\partial}{\partial t}\phi_s^*(e^{f_t}\alpha_s) = \phi_s^*(\frac{\partial}{\partial t}e^{f_t}\alpha_s) = \phi_s^*(\dot{f}_t e^{f_t}\alpha_s)$$

And lastly

$$\frac{\partial}{\partial t}\phi_s^*(e^{f_s}\alpha_t) = \phi_s^*(e^{f_s}\frac{\partial}{\partial t}\alpha_t) = \phi_s^*(e^{f_s}\dot{\alpha}_t)$$

Finally plug in $s = t$ and sum the results:

$$\begin{aligned}\frac{d}{dt}\phi_t^*(e^{f_t}\alpha_t) &= \phi_t^*((X_t \cdot f_t)e^{f_t}\alpha_t + e^{f_t}\mathcal{L}_{X_t}\alpha_t) + \phi_t^*(\dot{f}_t e^{f_t}\alpha_t) + \phi_t^*(e^{f_t}\dot{\alpha}_t) \\ &= \phi_t^*(e^{f_t}((X_t \cdot f_t)\alpha_t + \mathcal{L}_{X_t}\alpha_t + \dot{f}_t\alpha_t + \dot{\alpha}_t))\end{aligned}$$

3 Algebraic Topology

3.1 General Statements about Homology

In the following let $(H_\bullet, \partial_\bullet)$ be any homology theory.

Lemma 3.1. *Let $A \subseteq X$ be a retract. Then $H_\bullet(X) \cong H_\bullet(X,A) \oplus H_\bullet(A)$.*

Proof. Let $r\colon X \to A$ be a retraction and $i\colon A \to X$ the inclusion. Then $H_\bullet(r) \circ H_\bullet(i) = \mathrm{id}_{H_\bullet(A)}$, in particular $H_\bullet(i)$ is injective. Hence the long exact sequence for the pair (X,A) breaks up into short split exact sequences

$$\begin{tikzcd} 0 \arrow[r] & H_n(A) \arrow[r] & H_n(X) \arrow[l, bend right, "H_n(r)"'] \arrow[r] & H_n(X,A) \arrow[r] & 0 \end{tikzcd}$$

and the splitting lemma yields the desired isomorphism. ◻

Lemma 3.2. *Let $A \subseteq X$ be a subspace, such that the inclusion $A \to X$ is a homotopy equivalence. Then $H_\bullet(X,A) = 0$.*

Proof. This follows from the long exact sequence for the pair (X,A). Note that the inclusion $(A,A) \to (X,A)$ is not necessarily a homotopy equivalence. ◻

Lemma 3.3. *Let $A \subseteq X$ be a closed neighbourhood-deformation retract. Then the projection $X \to X/A$ induces a natural isomorphism*

$$H_\bullet(X,A) \to H_\bullet(X/A,*)$$

.

Proof. Applying excision in the push-out

$$\begin{tikzcd} A \arrow[r] \arrow[d] & X \arrow[d] \\ * \arrow[r] & X/A \end{tikzcd}$$

shows that the projection induces an isomorphism. Furthermore, any map $(X,A) \to (Y,B)$ descends to a map $X/A \to Y/B$ and naturality then just follows from functoriality.

◻

3.2 CW-Complexes

Proposition 3.4. *Let (X, A) be a relative CW-complex. If A is Hausdorff (resp. normal) then X is Hausdorff (resp. normal) as well. In particular every absolute CW-complex is Hausdorff and normal.*

Proof. We show the statement for the case that A is normal. The Hausdorff case follows completely analogously by replacing the sets to be seperated by point sets. So let $B, C \subseteq X$ be closed disjoint sets. We will show the following claim inductively:

Claim. There exist open disjoint sets $U_n, V_n \subseteq X_n$, such that $B_n := B \cap X_n \subseteq U_n$ and $C_n := C \cap X_n \subseteq V_n$ as well as $U_{n-1} \subseteq U_n, V_{n-1} \subseteq V_n$.

The case $n = -1$ follows from the assumption on A. For the induction step choose a push-out

$$\begin{tikzcd}
\coprod_{i\in I} S^{n-1} \arrow[r, "(q_i)"] \arrow[d] & X_{n-1} \arrow[d] \\
\coprod_{i\in I} D^{n} \arrow[r, "(Q_i)"] & X_n
\end{tikzcd}$$

and set $U'_{n-1} := (q_i)^{-1}_{i\in I}(U_{n-1})$, $V'_{n-1} := (q_i)^{-1}_{i\in I}(V_{n-1})$. By Lemma 3.5 shown below we can find open neighbourhoods $U'_n, V'_n \subseteq \coprod_I D^n$ with the following properties:

(i) $(Q_i)_I^{-1}(B_n) \subseteq U'_n$ and $(Q_i)_I^{-1}(C_n) \subseteq V'_n$

(ii) $U'_n \cap V'_n = \emptyset$

(iii) $U'_{n-1} = U'_n \cap \coprod_I S^{n-1}$ and $V'_{n-1} = V'_n \cap \coprod_I S^{n-1}$

Then define

$$U_n := U_{n-1} \cup (Q_i)_I(U'_n), \quad V_n := V_{n-1} \cup (Q_i)_I(V'_n)$$

By (iii) $U_n \cap X_{n-1} = U_{n-1}$ and $(Q_i)_I^{-1}(U_n) = U'_n$, so $U_n \subseteq X_n$ is open. The same holds for V_n. Furthermore, $B_n \subseteq U_n$, $C_n \subseteq V_n$ by (i) and $U_n \cap V_n = \emptyset$ by (ii). This shows the claim. Now consider

$$U := \bigcup_n U_n, \quad V := \bigcup_n V_n.$$

By construction $U \cap X_n \subseteq$ is open for all $n \geq -1$ and hence $U \subseteq X$ is open by definition of the colimit topology. The same holds for V. Moreover, $B \subseteq U, C \subseteq V$ and $U \cap V = \emptyset$. $\square$

It remains to proof the lemma mentioned in the proof of the above proposition.

Lemma 3.5. *Let $B, C \subseteq D^n$ be closed disjoint sets and $U', V' \subseteq S^{n-1}$ open disjoint neighbourhoods of $B \cap S^{n-1}, C \cap S^{n-1}$, respectively. Then there exist open sets $U, V \subseteq D^n$ with the following properties:*

(i) $B \subseteq U, C \subseteq V$

(ii) $U \cap V = \emptyset$

(iii) $U' = U \cap S^{n-1}, V' = V \cap S^{n-1}$

Proof. Consider the open disjoint sets

$$U'' := \{rx \mid x \in U',\, 0 < r \leq 1\}$$

and

$$V'' := \{rx \mid x \in V',\, 0 < r \leq 1\}.$$

Moreover, since B and $\overline{V}'$ are disjoint and closed we can find an open neighbourhood W_B of B, such that

$$\overline{W}_B \cap \overline{V}' = \emptyset.$$

Similialry we can find an open neighbourhood W_C of C with

$$\overline{W}_C \cap \overline{U}' = \emptyset.$$

After potentially shrinking W_B and W_C we may also assume, that they are disjoint. Now set

$$U := (U'' - \overline{W}_C) \cup (W_B - S^{n-1})$$

and

$$V := (V'' - \overline{W}_B) \cup (W_C - S^{n-1}).$$

It is then easy to verify, that these sets satisfy the statement. $\square$

3.3 Remarks about Urs' lecture on Algebraic Topology

In this section we present an alternative (and arguably more elegant) proof on how to extend the fact that de-Rham cohomology and singular cohomology of open subsets of $\mathbb{R}^n$ are isomorphic to manifolds with or without boundary, while at the same time formulating the statement in more category theoretical terms.

Lemma 3.6. *The map*

$$\Gamma\colon \Omega^k(M) \to C_k^\infty(M)^*$$
$$\omega \mapsto (\sigma \mapsto \int_\sigma \omega := \int_{\Delta^k} \sigma^*\omega)$$

defines a natural map of cochain complexes

$$(\Omega^\bullet(M), d) \to (C_k^\infty(M)^*, \partial^*),$$

i.e. a natural transformation of functors from the category of smooth manifolds to the category of cochain complexes over $\mathbb{R}$.

Proof. For smooth manifolds M, N and a smooth map $f\colon M \to N$ we have to show the commutativity of the following diagrams:

$$\begin{array}{ccc} \Omega^k(M) & \xrightarrow{\Gamma} & C_k^\infty(M)^* \\ \Big\downarrow d & & \Big\downarrow \partial^* \\ \Omega^{k+1}(M) & \xrightarrow{\Gamma} & C_{k+1}^\infty(M)^* \end{array} \qquad \begin{array}{ccc} \Omega^k(N) & \xrightarrow{\Gamma} & C_k^\infty(N)^* \\ \Big\downarrow f^* & & \Big\downarrow f^* \\ \Omega^k(M) & \xrightarrow{\Gamma} & C_k^\infty(M)^* \end{array}$$

For the left diagram we compute:

$$(\Gamma \circ d)(\omega) = \left(\sigma \mapsto \int_\sigma d\omega\right) = \left(\sigma \mapsto \int_{\partial\sigma} \omega\right) = (\partial^* \circ \Gamma)(\omega)$$

where we used Stokes' Theorem for singular chains in the second equality. Moreover, we compute for the right diagram:

$$\begin{aligned} (\Gamma \circ f^*)(\omega) &= \left(\sigma \mapsto \int_\sigma f^*\omega\right) \\ (f^* \circ \Omega)(\omega) &= \left(\sigma \mapsto \int_{f\circ\sigma} \omega\right) \end{aligned}$$

But by definition:

$$\int_\sigma f^*\omega = \int_{\Delta^k} \sigma^* f^* \omega = \int_{\Delta^k} (f \circ \sigma)^* \omega = \int_{f\circ\sigma} \omega$$

This completes the proof. $\square$

Lemma 3.7. *The inclusion $C_k^\infty(M) \to C_k(M)$ induces a natural map of cochain complexes*

$$(C_k^\infty(M)^*, \partial^*) \to (C_k(M)^*, \partial^*)$$

Proof. Obvious. $\square$

In the lecture we have shown that the cochain maps from the above two lemmata induce isomorphisms in homology at least for open subsets of $\mathbb{R}^n$. A short proof of how this generalizes to all manifolds with or without boundary is given in the following theorem (which notably uses facts that we have not established in the lecture).

Theorem 3.8. *The cochain maps from Lemma 3.6 and 3.7 induce isomorphisms in homology.*

Proof. $\square$

4 Fiber Bundles

4.1 Construction of fiber bundles

For convencience, let us state the definition of a fiber bundle.

Definition 4.1 (Fiber Bundle)**.** Let G be a topological group acting effectively on a space F. A fiber bundle E over B with fiber F and structure group G consists of a surjective continuous map $p\colon E \to B$ and a collection of maps

$$(\phi_\alpha\colon U_\alpha \times F \to p^{-1}(U_\alpha))_\alpha,$$

called a trivialising atlas[3], for open sets $U_\alpha \subseteq B$, such that

(i) The maps ϕ_α are homeomorphisms and the following diagrams commute:

$$\begin{tikzcd} U_\alpha \times F \arrow[rr, "\phi_\alpha"] \arrow[dr, "\pi"'] & & p^{-1}(U_\alpha) \arrow[dl, "p"] \\ & U_\alpha & \end{tikzcd}$$

where π is the usual projection onto the first factor.

(ii) For any two maps ϕ_α, ϕ_β there exists a continuous map

$$\tau_{\alpha,\beta}\colon U_\alpha \cap U_\beta \to G,$$

such that

$$\phi_\alpha^{-1} \circ \phi_\beta\colon (U_\alpha \cap U_\beta) \times F \to (U_\alpha \cap U_\beta) \times F$$

is given by $(b, f) \mapsto (b, \tau_{\alpha,\beta} \cdot f)$.

(iii) $(U_\alpha)_\alpha$ is an open cover of B.

(iv) The collection (ϕ_α) is maximal with respect to the first 3 conditions.

The maps ϕ_α are called local trivialisations, The sets U_α are called trivialising neighbourhoods and the map $\tau_{\alpha,\beta}$ is called the transition function for ϕ_α, ϕ_β. Furthermore, we call E the total space, B the base space and F the fiber. We write

$$F \hookrightarrow E \xrightarrow{p} B$$

to denote the fiber bundle.

Remark 4.2. We require surjectivity of p to exclude the case $F = \emptyset$. The requirement, that G acts effectively guarantees for the transition maps to be

[3]Wikipedia calls this a *G-Atlas*

unique. However, this is not a strong requirement to impose, since if G does not act effectively, i.e.

$$\Phi \colon G \to \mathrm{Aut}(F)$$

is not injective we get an effective group action by taking a quotient of G:

$$G/\mathrm{ker}\Phi \to \mathrm{Aut}(F).$$

It follows from the universal property of the quotient topology, that $G/\mathrm{ker}\Phi$ still acts continuously on F, i.e. the induced map $G/\mathrm{ker}\Phi \times F \to F$ is still continuous (if F is locally compact we can define the exponential topology on $\mathrm{Aut}(F)$ and the statement is equivalent to $G/\mathrm{ker}\Phi \to \mathrm{Aut}(F)$ being continuous).

Remark 4.3. The maximality condition for the trivialising atlas is well-defined: Given any trivialising atlas $\mathcal{F} = (\phi_\alpha)_\alpha$ satisfying conditions (i) - (iii) of Definition 4.1 we can define

$$\begin{aligned}\hat{\mathcal{F}} := \{\phi \colon U \times F \to p^{-1}(U) \mid U \subseteq B \text{ open }, \phi \text{ homeomorphism, for any } \phi_\alpha \in \mathcal{F} \\ \text{exists a map } \tau_{\alpha,\phi} \colon U \cap U_\alpha \to G, \text{ with } (\phi_\alpha^{-1} \circ \phi)(b, f) = (b, \tau_{\alpha,\phi}(b) \cdot f)\}.\end{aligned}$$

This is the maximal trivialising atlas induced by $\mathcal{F}$ satisfying (i) - (iii). Note however, that there is no unique trivialising atlas, since beginning with a different $\mathcal{F}$ may yield a different $\hat{\mathcal{F}}$ (as is the case for smooth atlases of smooth manifolds).

Proof of above remark. Given $\phi \colon U \times F \to p^{-1}(U), \phi' \colon U' \times F \to p^{-1}(U') \in \hat{\mathcal{F}}$, we claim that there exists $\tau \colon U \cap U' \to G$, such that

$$(\phi^{-1} \circ \phi')(b, f) = (b, \tau(b) \cdot f).$$

For $b_0 \in B$ choose $(\phi_\alpha \colon U_\alpha \times F \to p^{-1}(U_\alpha)) \in \mathcal{F}$ with $b_0 \in U_\alpha$. Then there exists

$$\tau_{\alpha,\phi} \colon U \cap U_\alpha \to G, \; \tau_{\alpha,\phi'} \colon U' \cap U_\alpha \to G$$

with

$$(\phi_\alpha^{-1} \circ \phi)(b, f) = (b, \tau_{\alpha,\phi}(b) \cdot f), \; (\phi_\alpha^{-1} \circ \phi')(b, f) = (b, \tau_{\alpha,\phi'}(b) \cdot f).$$

Then

$$(\phi^{-1} \circ \phi_\alpha) \circ (\phi_\alpha^{-1} \circ \phi') \colon (U_\alpha \cap U \cap U') \times F \to (U_\alpha \cap U \cap U') \times F$$

is given by

$$(b, f) \mapsto (b, \tau_{\alpha,\phi}(b)^{-1} \tau_{\alpha,\phi'}(b) \cdot f).$$

We conclude that around $\{b_0\} \times F$ we have

$$(\phi^{-1} \circ \phi')(b, f) = (b, \tau(b) \cdot f)$$

for $\tau(b) := \tau_{\alpha,\phi}(b)^{-1} \tau_{\alpha,\phi'}(b)$. Since group inversion and multiplication are continuous, τ is continuous. As b_0 was chosen arbitrary, such a map τ can be defined on all of $U \cap U'$. Furthermore, the group action of G is effective, hence τ is unique and the above argument shows that it is continuous. $\square$

For the transition functions we have the following Lemma, where again the effectiveness of the group action is crucial.

Lemma 4.4. *Let $F \hookrightarrow E \xrightarrow{p} B$ be a fiber bundle with structure group G. Then (using the notation in Definition 4.1):*

(i) $\tau_{\alpha,\alpha}(b) = e$

(ii) $\tau^{-1}_{\alpha,\beta}(b) = \tau_{\beta,\alpha}(b)$

(iii) $\tau_{\gamma,\alpha}(b)\tau_{\alpha,\beta}(b) = \tau_{\gamma,\beta}(b)$

Proof. Let us first proof (iii), then (i) and (ii) follow directly from the group axioms. We compute:

$$\begin{aligned}(b, \tau_{\gamma,\beta}(b) \cdot f) &= (\phi^{-1}_{\gamma} \circ \phi_{\beta})(b, f) \\ &= (\phi^{-1}_{\gamma} \circ \phi_{\alpha} \circ \phi^{-1}_{\alpha} \circ \phi_{\beta})(b, f) = (b, \tau_{\gamma,\alpha}(b)\tau_{\alpha,\beta}(b) \cdot f).\end{aligned}$$

Since the transition functions are uniquely determined, the statement follows. □

Interestingly, the transition functions already uniquely determine the fiber bundle, in a sense made precise by the following theorem.

Theorem 4.5 (Fiber bundle construction A)**.** *Let B, F be (non-empty) topological spaces and G a topological group acting effectively on F. Suppose we are given the following data:*

(i) An open cover $\mathcal{U} := (U_\alpha)_\alpha$ of B (where α ranges over some index set A)

(ii) A collection $(\tau_{\alpha,\beta})$ of continuous maps

$$\tau_{\alpha,\beta} \colon U_\alpha \cap U_\beta \to G$$

for each pair $U_\alpha, U_\beta \in \mathcal{U}$

(iii) $\tau_{\gamma,\alpha}(b)\tau_{\alpha,\beta}(b) = \tau_{\gamma,\beta}(b)$ for all $b \in U_\alpha \cap U_\beta \cap U_\gamma$

Then there exists a unique (up to isomorphism) fiber bundle $F \hookrightarrow E \xrightarrow{p} B$ with structure group G, such that the maximal trivialising atlas is induced by a trivialising atlas of the form

$$(\phi_\alpha \colon U_\alpha \times F \to p^{-1}(U_\alpha))_\alpha,$$

such that $(\phi^{-1}_{\alpha} \circ \phi_{\beta})(b, f) = (b, \tau_{\alpha,\beta}(b) \cdot f)$.

Proof. Consider the space

$$E := \left(\coprod_{\alpha \in A} U_\alpha \times F \right) \Big/ \sim$$

where the equivalence relation $\sim$ is defined by

$$(b, f, \beta) \sim (b, \tau_{\alpha,\beta}(b) \cdot f, \alpha)$$

and let $p\colon E \to B$, $[(b, f, \alpha)] \mapsto b$. Furthermore, let $\pi\colon \coprod_\alpha U_\alpha \times F \to E$ be the projection map.

Step 1. $\sim$ *is an equivalence relation.*

Transitivity follows directly from condition (iii). Reflexivity and symmetry then follow from the group axioms.

Step 2. p *is well-defined, continuous and surjective.*

Clearly p is well-defined and surjective as $F \neq \emptyset$. Continuity then follows directly from the universal property of the quotient topology.

Step 3. *Defining the trivialising atlas*

Consider

$$\phi_\alpha\colon U_\alpha \times F \to p^{-1}(U_\alpha),\ (b, f) \mapsto [(b, f, \alpha)].$$

Clearly ϕ_α is well-defined ($[(b, f, \alpha)] \in p^{-1}(U_\alpha)$) and continuous (the projection map π is continuous). The inverse is given by

$$\phi_\alpha^{-1}([(b, f, \beta)]) = (b, \tau_{\alpha,\beta}(b) \cdot f).$$

To show that ϕ_α^{-1} is continuous, we show that ϕ_α is an open map. To this end, let $U \times V \subseteq U_\alpha \times F$ be open and consider

$$\theta_{\alpha,\beta}\colon (U_\alpha \cap U_\beta) \times F \to (U_\alpha \cap U_\beta) \times F,\ (b, f) \mapsto (b, \tau_{\alpha,\beta}(b) \cdot f).$$

Clearly $\theta_{\alpha,\beta}$ is continuous (in fact it is a homeomorphism), hence

$$V_\beta := \theta_{\alpha,\beta}^{-1}((U \cap U_\beta) \times V) \subseteq (U_\alpha \cap U_\beta) \times F$$

is open and thus

$$\pi^{-1}(\phi_\alpha(U \times V)) = \coprod_{\beta \in A} V_\beta \subseteq \coprod_{\beta \in A} U_\beta \times F$$

is open. Thus $\phi_\alpha(U \times V)$ is open in E and since $p^{-1}(U_\alpha)$ is also open, we conclude that ϕ_α is an open map. Furthermore, the diagram

$$\begin{tikzcd} U_\alpha \times F \arrow[r, "\phi_\alpha"] \arrow[dr] & p^{-1}(U_\alpha) \arrow[d, "p"] \\ & U_\alpha \end{tikzcd}$$

commutes, such that ϕ_α is a local trivialisation. Moreover,

$$(\phi_\alpha^{-1} \circ \phi_\beta)(b, f) = (b, \tau_{\alpha,\beta}(b) \cdot f).$$

So the collection of maps $(\phi_\alpha)_\alpha$ is a trivialising atlas with the desired transition functions, making $F \hookrightarrow E \xrightarrow{p} B$ into a fiber bundle with structure group G.

Step 4. *Uniqueness*

Given two fiber bundles

$$\begin{aligned} \delta &:= (F \hookrightarrow E \xrightarrow{p} B) \\ \varepsilon &:= (F \hookrightarrow \tilde{E} \xrightarrow{\tilde{p}} B) \end{aligned}$$

with structure group G and trivialising atlases

$$\begin{aligned} &(\phi_\alpha \colon U_\alpha \times F \to p^{-1}(U_\alpha))_{\alpha \in A} \\ &(\tilde{\phi}_\alpha \colon U_\alpha \times F \to \tilde{p}^{-1}(U_\alpha))_{\alpha \in A} \end{aligned}$$

respectively, such that

$$(\phi_\alpha^{-1} \circ \phi_\beta)(b, f) = (b, \tau_{\alpha,\beta}(b) \cdot f) = (\tilde{\phi}_\alpha^{-1} \circ \tilde{\phi}_\beta)(b, f).$$

We can then construct an isomorphism $\Gamma \colon \delta \to \varepsilon$ as follows: for $x \in E$ choose $U_\alpha \in \mathcal{U}$ with $p(x) \in U_\alpha$. Then define

$$\Gamma(x) := (\tilde{\phi}_\alpha \circ \phi_\alpha^{-1})(x).$$

Well-definedness of Γ follows from the commutativity of the following diagram:

$$\begin{tikzcd} & (U_\alpha \cap U_\beta) \times F \arrow[dl, "\phi_\beta"'] \arrow[dr, "\tilde{\phi}_\beta"] \arrow[dd, "\tau"] & \\ p^{-1}(U_\alpha \cap U_\beta) & & \tilde{p}^{-1}(U_\alpha \cap U_\beta) \\ & (U_\alpha \cap U_\beta) \times F \arrow[ul, "\phi_\alpha"] \arrow[ur, "\tilde{\phi}_\alpha"'] & \end{tikzcd}$$

where $\tau(b, f) := (b, \tau_{\alpha,\beta}(b) \cdot f)$. Locally Γ is given by a continuous map, so it is continuous. Analogously we define the inverse map. Furthermore,

$$\begin{tikzcd} E \arrow[rr, "\Gamma"] \arrow[dr, "p"'] & & \tilde{E} \arrow[dl, "\tilde{p}"] \\ & B & \end{tikzcd}$$

commutes, so Γ is a bundle isomorphism. This completes the proof. $\square$

We may also formulate Theorem 4.5 slightly different, if we are given the fibers $F_x = p^{-1}(x)$ as sets. This is usually more practical, as the description of the bundle is more explicit:

Theorem 4.6 (Fiber bundle construction B)**.** *Let B, F be (non-empty) topological spaces and G a topological group acting effectively on F. Given sets F_x for each $x \in B$ we define*

$$E := \coprod_{x \in E} F_x$$

and $p\colon E \to B$ by sending each element in F_x to x. Suppose we are given the following data:

(i) An open cover $\mathcal{U} := (U_\alpha)_\alpha$ of B (where α ranges over some index set A)

(ii) A collection of bijections $\phi_\alpha\colon U_\alpha \times F \to p^{-1}(U_\alpha)$, that restrict to bijections $\{x\} \times F \to F_x$

(iii) A collection $(\tau_{\alpha,\beta})$ of continuous maps

$$\tau_{\alpha,\beta}\colon U_\alpha \cap U_\beta \to G$$

for each pair $U_\alpha, U_\beta \in \mathcal{U}$, such that $(\phi_\alpha^{-1} \circ \phi_\beta)(b, f) = (b, \tau_{\alpha,\beta}(b) \cdot f)$

Then E has a unique topology making $F \hookrightarrow E \xrightarrow{p} B$ into a fiber bundle with structure group G, where the maps ϕ_α are local trivialisations.

Proof. Define the topology on E as the final topology with respect to the maps ϕ_α, i.e.

$$V \subseteq E \text{ is open } \Leftrightarrow \phi_\alpha^{-1}(V) \subseteq U_\alpha \times F \text{ is open for each } \alpha$$

We show that the maps ϕ_α are open with this topology. To this end, let $W \subseteq U_\alpha \times F$ be open. Then

$$\phi_\beta^{-1}(\phi_\alpha(W)) = (\phi_\beta^{-1} \circ \phi_\alpha)(W \cap ((U_\alpha \cap U_\beta) \times F)) \subseteq U_\beta \times F$$

is open, since $\phi_\beta \circ \phi_\alpha$ is a homeomorphism by condition (iii). Thus ϕ_α is a homeomorphism. In particular, for any open set $U \subseteq U_\alpha$, we get that

$$p^{-1}(U) = \phi_\alpha(U \times F)$$

is open, since by (ii) the following diagram commutes:

$$\begin{tikzcd} U_\alpha \times F \arrow[rr, "\phi_\alpha"] \arrow[dr] & & p^{-1}(U_\alpha) \arrow[dl, "p"'] \\ & U_\alpha & \end{tikzcd}$$

So p is also continuous. Moreover, as $F \neq \emptyset$, we get that p is surjective. It remains to show uniqueness. So let E be equipped with any topology, such that p is continuous and ϕ_α are homeomorphisms. Let $V \subseteq E$, such that

$$W_\alpha := \phi_\alpha^{-1}(V) \subseteq U_\alpha \times F$$

is open for all α. Then
$$V = \bigcup_{\alpha} W_\alpha,$$
since the U_α cover B (and hence $E = \bigcup_\alpha \mathrm{im}\phi_\alpha$). As the ϕ_α are homeomorphisms and $p^{-1}(U_\alpha)$ is open, we conclude that V is open. Furthermore, the topology can not be finer than the final topology defined above, such that the two must coincide. □

Given a fiber bundle, Theorem 4.5 allows to change the fiber:

Corollary 4.7 (Changing the fiber)**.** *Let $F \hookrightarrow E \xrightarrow{p} B$ be a fiber bundle with structure group G. Given a space F' upon which G acts effectively, there exists a unique fiber bundle*
$$F' \hookrightarrow E' \xrightarrow{p'} B$$
which has the same trivialising neighbourhoods (U_α) and the same transition functions $(\tau_{\alpha,\beta})$ as $F \hookrightarrow E \xrightarrow{p} B$.

Theorem 4.5 tells us, that the new bundle still retains all of the information of the old bundle, as it is uniquely determined by the transition functions.

4.2 Smooth bundles

If we are in the smooth category, we can analogously define smooth fiber bundles by requiring the spaces E, B and F to be smooth manifolds (where either F or B has no boundary), the group G to be a Lie group and all maps to be smooth. In particular, the local trivialisation are diffeomorphisms. All of the above results also hold in the smooth category.

Theorem 4.8 (Fiber bundle construction A$'$)**.** *The conclusions of Theorem 4.5 remain true in the smooth category.*

Proof. The only difficulty is in defining the smooth structure on
$$E := \left(\coprod_{\alpha \in A} U_\alpha \times F \right) \Big/ \sim$$
This smooth structure is defined via the local trivialisations (ϕ_α) as follows: Choose and open cover (V_α) of B, such that

- $V_\alpha \subseteq U_\alpha$
- there exist smooth charts $h_\alpha \colon W_\alpha \to V_\alpha$

Furthermore, choose a smooth atlas
$$(k_i \colon W_i^F \to V_i^F)_i$$
of F. Then
$$(\phi_\alpha \circ (h_\alpha \times k_i) \colon W_\alpha \times W_i^F \to E)_{i,\alpha}$$

defines a smooth atlas for E, since the maps $\phi_\alpha^{-1} \circ \phi_\beta$ are diffeomorphisms. It is then easy to verify, that this atlas satisifies the first countability and Hausdorff axioms. Furthermore, this smooth structure is uniquely determined by requiring the local trivialisations to be diffeomorphisms. □

Theorem 4.9 (Fiber bundle construction B′)**.** *The conclusions of Theorem 4.6 remain true in the smooth category.*

Proof. As in the above theorem, the local trivialisations determine a unique smooth structure on E. □

4.3 Vector bundles

Whitney Sums

Given vector bundles E_1, E_2 over the same base B, we can define their **Whitney Sum** $E_1 \oplus E_2$ using Theorem 4.6 as the vector bundle whose fibers are given by $E_1|_p \oplus E_2|_p$ with the obvious choice as local trivialisations. The following Lemma characterizes the Whitney sum and at the same time provides the technical grounds for an intuition.

Lemma 4.10 (Characterisation of the Whitney sum)**.** *Let E_1 and E_2 be vector bundles over the same base space B of rank k_1 and k_2 respectively. A rank $k_1 + k_2$ vector bundle E over B is isomorphic to $E_1 \oplus E_2$ if and only if E_1 and E_2 can be embedded into E as subbundles and $E_1|_p + E_2|_p = E|_p$ for all $p \in B$.*

Let us first clarify what we mean by embedding. A vector bundle homomorphism $F \colon E \to E'$ is called and **embedding**, if it is injective. Then $\mathrm{im}(F)$ is a subbundle of E' and $F \colon E \to \mathrm{im}(F)$ is a vector bundle isomorphism. This follows from [Lee11] Theorem 10.34 and Prop. 10.26.

Proof of Lemma 4.10. First let $E = E_1 \oplus E_2$. Then we can define embeddings by

$$F_1 \colon E_1 \to E,\ v \mapsto (v, 0), F_2 \colon E_2 \to E\ w \mapsto (0, w).$$

Now let $E_1, E_2 \subseteq E$ be subbundles. Let $(U_\alpha)_\alpha$ be an open cover of B, such that E_1 and E_2 are trivial over U_α. Choose local frames

$$\sigma^i_{\alpha,1}, \dots, \sigma^i_{\alpha,k_i} \colon U_\alpha \to E_i$$

and define local trivialisations

$$\phi^i_\alpha \colon U_\alpha \times \mathbb{R}^{k_i} \to E_i,\ (p, v) \mapsto \sum_{j=1}^{k_i} v_j \sigma^i_{\alpha,j}(p).$$

Let $\tau^i_{\alpha,\beta} \colon U_\alpha \cap U_\beta \to \mathrm{GL}(k_i)$ be the transition functions defined by

$$((\phi^i_\alpha)^{-1} \circ \phi^i_\beta)(p, v) = (p, \tau^i_{\alpha,\beta}(p) \cdot v).$$

Since $E|_p$ is the direct sum of $E_1|_p$ and $E_2|_p$, we get that

$$\sigma^1_{\alpha,1}, \ldots, \sigma^1_{\alpha,k_1}, \sigma^2_{\alpha,1}, \ldots, \sigma^2_{\alpha,k_2}$$

is a local frame for E, thus defines a local trivialisation by

$$\psi_\alpha \colon U_\alpha \times \mathbb{R}^{k_1} \times \mathbb{R}^{k_2} \to E,\ (p, v, w) \mapsto \phi^1_\alpha(p, v) + \phi^2_\alpha(p, w).$$

Then

$$(\psi_\alpha^{-1} \circ \psi_\beta)(p, v, w) = (p, \tau^1_{\alpha,\beta}(p) \cdot v, \tau^2_{\alpha,\beta}(p) \cdot w).$$

Note that $E_1 \oplus E_2$ is also trivial over U_α via

$$\varphi_\alpha \colon U_\alpha \times \mathbb{R}^{k_1} \times \mathbb{R}^{k_2} \to E_1 \oplus E_2,\ (p, v, w) \mapsto (\phi^1_\alpha(p, v), \phi^2_\alpha(p, w))$$

and $\varphi_\alpha^{-1} \circ \varphi_\beta = \psi_\alpha^{-1} \circ \psi_\beta$, thus

$$E \cong E_1 \oplus E_2$$

by the uniqueness result in Theorem 4.5. $\square$

This characterisation gives an intuitive reasoning behind the fact, that the Whitney sum of two Moebius bundles is trivial:

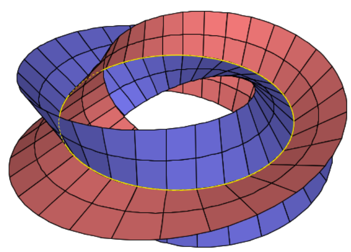

Figure 2: Illustration of the Whitney sum of two Moebius bundles [source: `https://math.stackexchange.com/questions/1009126/global-trivialization-of-m-oplus-m`]

Lemma 4.11. *Let E, E_1 and E_2 be vector bundles over a manifold M and $\sigma_i \colon M \to E_i$ smooth sections for $i = 1, 2$. Then the following maps are smooth:*

(i) $(\sigma_1, \sigma_2) \colon N \to E_1 \oplus E_2,\ p \mapsto (\sigma_1(p), \sigma_2(p))$

(ii) The addition map $E \oplus E \to E,\ (v, w) \mapsto v + w$

(iii) The scalar multiplication map $\mathbb{R} \times E \to E,\ (\lambda, v) \mapsto \lambda v$

Proof. All statements follow easily by writing the maps in local trivialisations. $\square$

The Normal Bundle

We can generalize Lemma 10.35 from [Lee11] to Riemannian manifolds:

Lemma 4.12 (Orthogonal complement bundles)**.** *Let M be a Riemannian manifold, $S \subseteq M$ a submanifold and $E \subseteq TM|_S$ a subbundle. Then*

$$E^\perp := \{(p, v) \mid p \in S, v \in E_p^\perp\} \subseteq TM|_S$$

is a subbundle and there exists a smooth orthonormal local frame for $E^\perp$ around each point in S.

Proof. Choose any local frame

$$\sigma_1, \ldots, \sigma_k \colon U \to E.$$

After possibly shrinking U, we can extend this frame to a frame

$$\sigma_1, \ldots, \sigma_n \colon U \to TM|_S$$

of $TM|_S$. Write $\sigma_i(p) = (p, X_p^i)$. Applying Gramm-Schmidt to the vectors X_p^i yields an orthonormal basis $Y_p^1, \ldots, Y_p^n$ of T_pM for each $p \in U$:

$$Y_p^i := \frac{X_p^i - \sum_{j=1}^{i-1} \langle X_p^i, Y_p^j \rangle X_p^i}{\| X_p^i - \sum_{j=1}^{i-1} \langle X_p^i, Y_p^j \rangle X_p^i \|}$$

Since this formula is smooth, we can define the smooth sections

$$\tilde{\sigma}_i(p) := (p, Y_p^i).$$

Notice that $\tilde{\sigma}_i(p) \in E^\perp$ for $i \geq k+1$ and $E_p^\perp = \operatorname{span}\{Y_p^{k+1}, \ldots, Y_p^n\}$. This shows the statement. $\square$

Given a Riemannian manifold M and a submanifold $S \subseteq M$, we can define the **normal bundle** of S:

$$NS := \{(p, v) \in TM \mid p \in S, v \in T_pS^\perp\}.$$

Corollary 4.13. *Let M be a Riemannian manifold, $S \subseteq M$ a submanifold. Then NS is a subbundle of $TM|_S$.*

Proof. We only need to show, that TS is a smooth subbundle of $TM|_S$ and then apply Lemma 4.12. To this end, choose a local frame (σ_i) of TS. Then $\iota \circ \sigma_i$ are sections of $TM|_S$, where $\iota \colon TS \hookrightarrow TM|_S$ is the inclusion. So we only have to verify, that ι is smooth. But this can easily be seen in local trivialisations of TS and $TM|_S$. $\square$

4.4 Pullback Bundles

Given a fiber bundle $F \hookrightarrow E \xrightarrow{p} B$ and a map $g\colon B' \to B$ we can construct the pullback bundle g^*E using Theorem 4.6 as the fiber bundle over B', where the fiber over $b' \in B'$ is given by the fiber of E over $g(b')$, i.e.

$$g^*E = \coprod_{b' \in B'} p^{-1}(g(b')).$$

Given a local trivialisation $\phi\colon U \times F \to E$ we define a local trivialisation of g^*E by

$$g^{-1}(U) \times F \to g^*E,\ (b', f) \mapsto (b', \phi(g(b'), f)).$$

The same construction works in the smooth category using Theorem 4.9 as long as either F has no boundary or B and B' have no boundary. By projecting onto the second factor we get a bundle homomorphism $G\colon g^*E \to E$, making the following diagram commute:

$$\begin{tikzcd} g^*E \arrow[r, "G"] \arrow[d] & E \arrow[d, "p"] \\ B' \arrow[r, "g"] & B \end{tikzcd}$$

Lemma 4.14. *In the above situation the following holds:*

(i) If g is an immersion, so is G

(ii) If g is an embedding, so is G

Proof. In local trivialisations we can write G as

$$g^{-1}(U) \times F \to U \times F,\ (b, f) \mapsto (g(b'), f),$$

such that (i) follows. Now let g be an embedding. We have $\mathrm{im}G = p^{-1}(g(B'))$, so we can define

$$H\colon \mathrm{im}G \to f^*E,\ x \mapsto (g^{-1}(p(x)), x).$$

Clearly H is continuous and $G \circ H = \mathrm{id}_{\mathrm{im}G}$ as well as $H \circ G = \mathrm{id}_{g^*E}$, so G is in fact an embedding. $\square$

Given a fiber bundle $F \hookrightarrow E \xrightarrow{p} B$ and a submanifold $S \subseteq B$ we can then define the restriction of E to S as the pullback of E under the inclusion $\iota\colon S \hookrightarrow B$:

$$E|_S := \iota^*E$$

The above Lemma shows that this is indeed a submanfold of E. In fact, in the situation of the above Lemma we have

$$\mathrm{im}G = E|_{g(B')}$$

5 Miscellaneous

5.1 Smoothness on arbitrary subsets

Let M, N be manifolds and $A \subseteq M$ an arbitrary subset. In [Lee11] the author defines what it means for a map

$$f\colon A \to N$$

to be smooth, namely

We call f smooth if around each $x \in A$ there exists a smooth extension $\widetilde{f}\colon U \to N$ of f to some open neighbourhood $U \subseteq M$ of x (see [Lee11, p. 45]).

However this definition has a fatal flaw: if A happens to be a submanifold of M, a map $f\colon A \to N$ might be smooth in the usual sense (i.e. as a map between smooth manifolds) but not smooth in the sense of Lee‘s definition. Consider for example

Counterexample 5.1.

$$M := \mathbb{R}, N := [0,1], A := [0,1],$$
$$f\colon A \to N,\, t \mapsto t.$$

Then clearly f is smooth as a map between smooth manifolds, but around $0, 1 \in A$ it does not admit a smooth extension onto an open neighbourhood $U \subseteq M$, since the image of such an extension would fail to be contained within N.

We can however correct this flaw by defining:

Definition 5.2 (Extension-smooth)**.** Let M^n, N^m be manifolds with or without boundary, $A \subseteq M$ an arbitrary subset and $f\colon A \to N$ a map. We call f *extension-smooth* if around each $x \in A$ there exist charts

$$h\colon U \to M \text{ around } x$$
$$k\colon V \to N \text{ around } f(x),$$

such that

$$g := k^{-1} \circ f \circ h|_{h^{-1}(A)}$$

admits a smooth extension around $h^{-1}(x)$ in the sense that there exist open subsets $U_0 \subseteq \mathbb{R}^n$ and $V_0 \subseteq \mathbb{R}^m$ with $h^{-1}(x) \in U_0$ and $k^{-1}(f(x)) \in V_0$ and a smooth map

$$\widetilde{g}\colon U_0 \to V_0$$

with $g|_{U_0 \cap h^{-1}(A)} = \widetilde{g}\, |_{U_0 \cap h^{-1}(A)}$.

We now claim the following.

Lemma 5.3. *If $\partial N = \emptyset$ then Lee's definition and Definition 5.2 are equivalent. In general, if f is smooth in the sense of Lee's definition, then it is also extension-smooth but the converse may fail.*

Lemma 5.4. *If A is a submanifold of M, then $f\colon A \to N$ is smooth as a map of smooth manifolds if and only if it is extension-smooth.*

In particular the extension Lemma in [Lee11] (Lemma 2.26.) still holds and the proof of the Whitney approximation theorems in chapter 6 work the same (here the extension Lemma is used). But with our new definition Problem 6-7 in [Lee11] now is an actual counterexample. (with Lee's definition F would not be smooth on the closed subset A but only smooth as a map between manifolds).

Another way to properly define smoothness of $f\colon A \to N$ would be to change the codomain in Lee's definition to be the double of N. This should be equivalent to Definition 5.2 and is arguably more elegant at the cost of being less elementary.

Smooth Homotopies

This allows us to define what a smooth homotopy is:

Definition 5.5. Let M, N be manifolds with boundary. A homotopy

$$H\colon M \times [0,1] \to N,$$

is called smooth, if H is extension-smooth (see Definition 5.2), where we consider $M \times [0,1]$ as a subset of the manifold $M \times \mathbb{R}$.

In [Lee11] the author defines a smooth homotopy as a map $H\colon M \times [0,1] \to N$, which can be extended to a smooth map $\overline{H}\colon M \times \mathbb{R} \to N$. As we have seen in Counterexample 5.1 this is a stronger requirement compared to Definition 5.5, as less homotopies are considered to be smooth by Lee's definition. However the homotopy classes of smooth maps are the same. More precisely this is formulated in the following lemma.

Lemma 5.6. *Let M, N be manifolds with or without boundary and $f, g\colon M \to N$ smooth maps. Then f and g are smoothly homotopic in the sense of Definition 5.5 if and only if there exists a homotopy $H\colon M \times [0,1] \to N$ between f and g, which can be extended to a smooth map $\overline{H}\colon M \times \mathbb{R} \to N$.*

Proof. Let $G\colon M \times [0,1] \to N$ be a smooth homotopy from f to g in the sense of Definition 5.5. Choose a smooth function $\phi\colon \mathbb{R} \to [0,1]$ with

$$\phi(t) = 0 \text{ for } t \leq 0$$
$$\phi(t) = 1 \text{ for } t \geq 1$$

Then

$$\Phi\colon M \times [0,1] \to M \times [0,1], (x,t) \mapsto (x, \phi(t))$$

is extension-smooth (it can be smoothly extended to $M \times \mathbb{R}$ and this implies that it is smooth in the sense of Definition 5.2, as Lee's definition is stronger by Lemma 5.3). Furthermore, it is easily verified that the composition of extension-smooth maps on arbitray subsets is extension-smooth, such that

$$H := G \circ \Phi \colon M \times [0,1] \to N$$

is extension-smooth. Then H is a homotopy from f to g and we can extend H to the smooth map

$$\overline{H} \colon M \times \mathbb{R} \to N, (x,t) \mapsto G(x, \phi(t)).$$

The converse follows directly from Lemma 5.3.

$\square$

6 Solutions to selected exercises

Exercise 6.1 (Exercise 10-10 from [Lee11])**.** Let M be a manifold without boundary of dimension n and $E \to M$ a rank k vector bundle over M. Then there exists a section $\sigma\colon M \to E$ with the following properties:

1. If $k > n$, then σ is nowhere vanishing
2. If $k \leq n$, then the zeros of σ form a closed codimension k submanifold of M

In particular, there exists a smooth vector field X on M, where the zeros of X are a discrete closed subset. Hence, every compact manifold with or without boundary admits a smooth vector field with only finitely many zeros.

Proof. Consider the zero section $\xi\colon M \to E$ and let $M' := \operatorname{im}(\xi)$. Then M' is a codimension 0 submanifold (without boundary) of E. By the transversality homotopy theorem (Theorem 1.5) any section $M \to E$ is homotopic to a section $\sigma\colon M \to E$ that is transversal to M'. If $k > n$ this is equivalent to

$$\sigma(M) \cap M' = \emptyset,$$

which in turn is equivalent to $\sigma(p) \neq 0$ for all $p \in M$. If $k \leq M$, then

$$\{p \in M \mid \sigma(p) = 0\} = \sigma^{-1}(M')$$

and $\sigma^{-1}(M')$ is a closed codimension k submanifold of M. This shows the first two statements. Applying these results to $E := TM$, we see that the zeros of σ are a closed codimension 0 submanifold, thus discrete. If M is compact this is equivalent to $\{p \in M \mid \sigma(p) = 0\}$ being finite. If M has non-empty boundary, we can construct a vector field with finitely many zeros on the double and restrict it to M. ☐

Exercise 6.2 (Own exercise)**.** Let M be the Moebius line bundle. Consider the map $f\colon S^1 \to S^1$, $z \mapsto z^2$. Then the pullback bundle $E := f^*M$ is trivial.

Intuitively, we can see this as follows: The fiber over $e^{\pi i}$ of E is just the fiber over $e^{2\pi i}$ of M, which has already made a half-twist, i.e. the fiber of E over 1 can be imagined as a straight line and then begins twisting upon increasing the angle. When the angle $\theta = \pi$ is reached a half twist has been performed. But upon further increasing the angle, another half twist is performed as we "walk" around the moebius bundle a second time. So E is basically a double twisted moebius bundle, which is trivial.

Proof. Proof of Exercise 6.2 We define $M := \mathbb{R} \times \mathbb{R}/_\sim$, where the equivalence relation is given by

$$(x, t) \sim (x + k, (-1)^k t)$$

and the projection $p\colon M \to S^1$ is defined by $p([(x,t)]) := e^{2\pi i x}$. By identifiying S^1 with $[0,1]/_{0\sim 1}$ we can write f as

$$f([\theta]) = e^{2\pi i\theta},$$

such that E is a line bundle over $[0,1]/_\sim$. To see that E is trivial we define a global frame:

$$\sigma\colon [0,1]/_\sim \to E,\ [\theta] \mapsto ([\theta], [(2\theta, 1)])$$

Note that $[(0,1)] = [(2,1)]$ in M, such that σ is a well-defined section. Furthermore, it is easily seen that this section is nowhere vanishing. To see that σ is smooth we could write σ in the local trivialisations of E. Alternatively we can argue as follows: Consider

$$s\colon [0,1] \to M,\ \theta \mapsto [(2\theta, 1)]$$

and let $\pi\colon [0,1] \to [0,1]/_\sim$ be the projection map. Then $s = \sigma \circ \pi$. Clearly s is smooth and it is easily verified that π is a local diffeomorphism, such that smoothness of σ follows. This completes the proof. $\square$

Exercise 6.3 (Exercise 10-14 from [Lee11])**.** Let M be a smooth manifold and $S \subseteq M$ a submanifold. Then TS is a subbundle of $TM|_S$.

Proof. Let $\sigma_1, \ldots, \sigma_k$ be a local frame for TS. Then we can also consider the σ_i as local sections of $TM|_S$ via the inclusion $TS \hookrightarrow TM|_S$. If we can show that these are still smooth sections we are done as this shows that TS (as a set) is a subbundle of $TM|_S$ an the induced smoothness structure is the same as the usual one on TS (see the construction of the smoothness structure on the subbundle in Lemma 10.32 in [Lee11] to see that this indeed yields the same structures). But smoothness of σ_i as local sections of $TM|_S$ follows directly from the fact that the inclusion $TS \hookrightarrow TM|_S$ is just the restriction of the differential of the inclusion $\iota\colon S \hookrightarrow M$ to the submanifold $TM|_S \subseteq TM$ in the codomain (the fact that $TM|_S$ is a submanifold of TM follows for example from Lemma 4.14). $\square$

Exercise 6.4 (Mentioned in [Bre13] in chapter 'Conventions for CW complexes')**.** Let (X, x_0) and (Y, y_0) be compact pointed topological spaces, then the smash product $X \wedge Y$ is homeomorphic to the one point compactification of $Z := (X - x_0) \times (Y - y_0)$. In particular

$$S^n \wedge S^m \cong S^{n+m}.$$

Proof sketch. Let Z^+ be the one point compactification of Z. Define $f\colon X \times Y \to Z^+$ by

$$f(x,y) := \begin{cases} (x,y) & , x \neq x_0 \text{ and } y \neq y_0 \\ \infty & , \text{otherwise} \end{cases}$$

It is then straightforward to show that f is continuous and the following diagram is a push-out:

$$\begin{tikzcd} X \vee Y \arrow[r] \arrow[d] & X \times Y \arrow[d, "f"] \\ * \arrow[r] & Z^+ \end{tikzcd}$$

So by the universal property of push-outs we get $Z^+ \cong X \wedge Y$. The statement about the smash product of spheres then just follows from $S^n - * \cong \mathbb{R}^n$ and that the one point compactification of $\mathbb{R}^n$ is S^n. $\square$

7 Alternative Proofs

This is a collection of alternative proofs for various statements from books, lecture notes etc.

Theorem 7.1 (Theorem 10.34. from [Lee11])**.** *Let E and E' be smooth vector bundles over a manifold M, and let $F\colon E \to E'$ be a bundle homomorphism. Then* $\ker F$ *and* $\operatorname{im} F$ *are smooth subbundles of E and E', respectively, if and only if F has constant rank.*

Proof. We present an alternative proof to show that $\ker F$ is a smooth subbundle. Let $\operatorname{rank} F = k$. Choose a local frame

$$\sigma_1, \dots, \sigma_n \colon U \to E,$$

such that $F \circ \sigma_1, \dots, F \circ \sigma_k$ are linearly independent sections, i.e. a local frame for $\operatorname{im} F$. So for $j > k$ we can write

$$F \circ \sigma_j = \sum_{i=1}^{k} \tau_{ij} F \circ \sigma_i$$

for smooth functions

$$\tau_{ij} \colon U \to \mathbb{R}$$

Define

$$\sigma_j' := \sigma_j - \sum_{i=1}^{k} \tau_{ij} \sigma_i$$

Then $F \circ \sigma_j' = 0$, thus $\operatorname{im} \sigma_j' \subseteq \ker F$. Furthermore, $\sigma_{k+1}', \dots, \sigma_n'$ are linearly independent, since $\sigma_1, \dots, \sigma_n$ is a local frame. This shows the statement. $\square$

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